Useful Memory Has a Horizon

Structural Limits of Drift Tracking

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Abstract

Under drift, memory does not stay equally useful: retaining more past information reduces statistical noise, but also increases temporal misalignment with the current distribution. As the target law moves, each additional sample becomes useful and increasingly stale, so the problem is not simply how much to remember, but how long remembered evidence should remain operationally current.

We study this trade-off under Wasserstein nonstationarity. Cube-root scalings already appear in nonstationary optimization and bandit literature, but here the object is the finite-memory horizon itself. For the finite-memory averaging class analyzed here, the tracking error splits into a statistical term that decreases with memory and a worst-case staleness term that grows with drift. When the statistical fluctuations scale as $n^{-1/2}$, as in the low effective-dimension regime for raw W_2 and in fixed-ε Sinkhorn-based measurement carriers, the resulting worst-case **cube-root law** holds: the **optimal memory horizon** scales as $\boldsymbol{n^* \asymp (C_K/\zeta)^{2/3}}$, while the corresponding **minimum tracking error** scales as $\boldsymbol{\mathcal{E}_{\min} \asymp C_K^{2/3}\zeta^{1/3}}$. In the critical-window regime, we also prove a minimax lower bound, showing that this finite-memory floor is structural rather than estimator-specific.

Empirically, Gaussian drift experiments recover the predicted U-curve and a finite useful-memory optimum. Continuous-drift benchmarks expose a gap between temporal validity and changepoint evidence; alternating-timescales benchmarks show that contraction is easier than re-expansion; and gap-cost experiments show that horizon misalignment induces nonlinear predictive cost.

The resulting picture is operational: useful memory under drift has geometry, temporal coherence is not the same object as changepoint evidence, and the predictive value of horizon control depends on the interaction between drift geometry and memory policy.

Keywords: tracking under drift; useful memory; temporal coherence; Wasserstein distance; Sinkhorn divergence; concept drift; adaptive memory.
MSC 2020: 93D05, 93D30, 68P99.

1 Introduction

Adaptive systems rarely track a fixed target. In streaming estimation, monitoring, and online decision-making, the data-generating distribution itself drifts over time. A learner with finite memory must therefore solve a basic but easy-to-miss problem: how much of the past should still count once the world that produced it has started to move away?

That question creates a structural trade-off. Using more past data reduces sampling noise, but it also increases the age of the information being used to track the present. Under drift, memory is simultaneously helpful and harmful: short windows are fresh but noisy, while long windows are stable but stale. The constraint is temporal rather than parametric: one cannot overparameterize time. The central claim here is that this tension defines a coherence-lag trade-off and induces a quantitative law for finite-memory tracking under Wasserstein nonstationarity. The cube-root exponents themselves are not unique; the finite-memory horizon interpretation and the temporal-validity floor are the new pieces here.

For the finite-memory averaging class analyzed here, the tracking error splits into a statistical term that decreases with memory length and a staleness term that increases with drift and effective data age. In the worst-case linear staleness regime induced by the W_2-Lipschitz path bound, optimizing that balance yields a cube-root law: $\mathcal{E}_{\min} \asymp C_K^{2/3}\zeta^{1/3}$ with $n^* \asymp (C_K/\zeta)^{2/3}$. The law is exact for the averaging class considered here. Beyond that class, the paper proves a minimax lower bound for window-restricted estimators in the critical-window regime, establishing that the finite-memory floor is structural rather than a quirk of one specific rule.

The cube-root scaling corresponds to the carrier regime where the finite-sample term decays as $n^{-1/2}$; more generally, carrier-dependent exponents lead to carrier-dependent horizon scaling remark 2.3. The analysis below is stated for the linear-staleness regime induced by the local W_2-Lipschitz path bound.

The same decomposition also has a simple interpretation: the limiting factor is the age of the information that finite memory forces the estimator to retain, alongside the usual finite-sample cost. The learner pays for variance reduction by carrying older evidence, and optimal tracking occurs at a memory horizon that balances statistical stability against operational staleness.

Horizon geometry, horizon tracking, and predictive payoff are not the same object. A regulator can be geometrically right and still be operationally limited; it can also preserve temporal validity without dominating every backend. The empirical sequence asks when the memory horizon is locally right, when it becomes stale, and how much that staleness costs.

Every backend need not collapse to the same horizon. Short fixed windows can remain strong compromises even when the local oracle horizon moves over time. What the UMR horizon cap contributes is a notion of validity: it tells us when remembered evidence is still current enough to matter, and when it has become stale before a detector has enough evidence to complain.

The empirical section follows that logic in stages. The first checks recover the 'U'-curve and show that the finite-memory optimum is not a fitting artifact. The next checks separate temporal validity from changepoint evidence, and the alternating-timescales benchmark shows that contraction is easier to induce than re-expansion. The gap-cost analysis then asks how a mismatch between the chosen horizon and the local oracle translates into predictive loss.

ELEC2 stays in the main text as the representative real-stream case, while a short Bikes summary moves to the appendix. The controlled downstream check in Appendix F adds a synthetic classification test of the same cap-only logic. The order is geometry first, temporal validity second, predictive payoff third.

The theoretical results are stated in W_2; the Sinkhorn divergence is the computational realization used in the synthetic measurement layer. In the public-stream arena, a secondary effect remains visible: EMA-based horizon recovery is asymmetric, with contraction after bursts faster than re-expansion after low-drift stretches.

The window size is a fundamental parameter whose mischoice can create spurious or missed detections Hinder et al. [2023]; the coherence-lag law provides a principled answer when the drift rate is estimable.

Finite-memory tracking under drift raises a simple question: what tracking accuracy is possible when the learner's memory is finite and the target distribution keeps moving?

2 Useful Memory Geometry

Under continuous drift, memory is not only a statistical resource but also a temporal liability. The core question is therefore not whether one can react to change, but how long a sample should remain operationally useful before it becomes stale.

Theorem 2.1 (Lag-Inclusive Tracking Error Bound)**.** *Let $X_{t-j} \sim P^*_{t-j}$ and let*

$$\widehat{P}_t^{(n)} \;=\; \frac{1}{n}\sum_{j=0}^{n-1} \delta_{X_{t-j}}$$

be the uniform-window empirical estimator. Assume the target path is locally W_2-Lipschitz with rate ζ, so that

$$W_2\big(P^*_{t-j}, P^*_t\big) \le \zeta j \qquad \text{for all } 0 \le j < n,$$

and assume the finite-sample term obeys

$$\mathbb{E}\Big[W_2\Big(\widehat{P}_t^{(n)}, \bar{P}_t^{(n)}\Big)\Big] \le C_K n^{-1/2},$$

where $\bar{P}_t^{(n)} := n^{-1}\sum_{j=0}^{n-1} P^*_{t-j}$. Then the expected tracking error satisfies

$$\mathcal{E}(n) \;\le\; C_K\, n^{-1/2} \;+ \tfrac{1}{2}\,\zeta\, n,$$

where the first term is the finite-sample statistical cost and the second term is the staleness cost induced by retaining old information.

Lemma 2.2 (Finite-sample Wasserstein term)**.** *Let Q be a probability measure on $\mathbb{R}^{d_{\text{eff}}}$ with support contained in a ball of radius R and finite second moment. Let $X_1, \ldots, X_n$ be i.i.d. samples from Q, and let $\hat{Q}_n := n^{-1}\sum_{i=1}^n \delta_{X_i}$. Then:*

- *If $d_{\text{eff}} < 4$, there exists a constant $C_{d_{\text{eff}},R}$ such that*

$$\mathbb{E}\, W_2(\hat{Q}_n, Q) \le C_{d_{\text{eff}},R}\, n^{-1/2}.$$

- *If $d_{\text{eff}} = 4$, the same bound holds with an additional $(\log n)^{1/2}$ factor.*
- *If $d_{\text{eff}} > 4$, the raw Wasserstein rate is slower than $n^{-1/2}$, so the constant-only form above should not be used.*

In that regime, the statistical fluctuations do not follow the $n^{-1/2}$ carrier exponent, so the resulting horizon scaling becomes carrier-dependent (see remark 2.3). If the geometry is computed with debiased Sinkhorn divergence S_ε on a bounded domain of diameter D, then for fixed $\varepsilon > 0$ the empirical fluctuations remain $O(n^{-1/2})$ with a constant $L(D, \varepsilon)$ that is dimension-free at fixed ε and increases as $\varepsilon \downarrow 0$.

A short numerical sanity check for this regime is reported in Appendix C; it is not used as a theorem, but it matches the intended low-dimensional interpretation of the finite-sample term.

Remark 2.3 (Carrier-dependent horizon scaling). The cube-root law follows when the finite-sample term behaves as $\mathbb{E}\, W_2(\widehat{P}_t^{(n)}, \bar{P}_t^{(n)}) \lesssim C_K n^{-1/2}$. More generally, if the finite-sample term admits the scaling

$$\mathbb{E}\, W_2(\widehat{P}_t^{(n)}, \bar{P}_t^{(n)}) \lesssim C_K n^{-a} \quad \text{for some } a > 0,$$

then balancing it against the drift-induced staleness cost $\frac{1}{2}\zeta n$ yields

$$n^*(a) \;=\; \left(\frac{2a\, C_K}{\zeta}\right)^{\frac{1}{a+1}}, \qquad \mathcal{E}_{\min}(a) \;=\; \frac{a+1}{(2a)^{a/(a+1)}}\, C_K^{1/(a+1)}\, \zeta^{a/(a+1)}.$$

In the setting of lemma 2.2, the effective exponent is $a = 1/2$ for $d_{\text{eff}} < 4$, with an additional $(\log n)^{1/2}$ factor for $d_{\text{eff}} = 4$; for $d_{\text{eff}} > 4$ the raw Wasserstein rate is slower than $n^{-1/2}$, so the exponent is strictly smaller than $1/2$ (carrier-dependent). The cube-root law is therefore the $a = 1/2$ *carrier instance* of a broader carrier-dependent horizon scaling.

The two terms in theorem 2.1 pull in opposite directions. Increasing n reduces variance but raises the age of the retained evidence. That tension is the basic coherence-lag trade-off that governs useful memory under drift.

Proposition 2.4 (Optimal Window Under Drift)**.** *Under theorem 2.1, the error $\mathcal{E}(n)$ is minimised at*

$$n^* \;=\; \left(\frac{C_K}{\zeta}\right)^{2/3},$$

with minimum value

$$\mathcal{E}_{\min} \;=\; \tfrac{3}{2}\, C_K^{2/3}\, \zeta^{1/3}.$$

Proof. Apply the triangle inequality for W_2:

$$W_2\Big(\widehat{P}_t^{(n)}, P_t^*\Big) \le W_2\Big(\widehat{P}_t^{(n)}, \bar{P}_t^{(n)}\Big) + W_2\Big(\bar{P}_t^{(n)}, P_t^*\Big).$$

The first term is controlled by the assumed finite-sample bound. For the second term, let π_j be an optimal coupling between P_{t-j}^* and P_t^*. Then the averaged coupling $\bar{\pi} := n^{-1} \sum_{j=0}^{n-1} \pi_j$ couples $\bar{P}_t^{(n)}$ with P_t^*, so the transport cost is at most the average of the individual costs. Combined with the W_2-Lipschitz drift bound, this gives

$$W_2\left(\bar{P}_t^{(n)}, P_t^*\right) \le \frac{1}{n}\sum_{j=0}^{n-1} W_2\left(P_{t-j}^*, P_t^*\right) \le \frac{1}{n}\sum_{j=0}^{n-1} \zeta j \le \tfrac{1}{2}\,\zeta n.$$

Combining the two bounds yields the stated inequality. □

Remark 2.5 (Measurement layer). The theorem is stated in W_2, so ε does not enter the mathematical drift law itself. In the synthetic measurement layer, however, the observable geometry is often a debiased Sinkhorn divergence with fixed ε, which makes the finite-sample constant calibration-dependent. In the synthetic benchmarks, $\hat{\zeta}_t$ is estimated from short-window block differences using an EMA update and C_K is calibrated from a stationary prefix; Appendix D records the sensitivity check and the prequential calibration rule.

For fixed ε on bounded domains, the debiased Sinkhorn divergence converges at rate $n^{-1/2}$ with ε-dependent constants [Genevay et al., 2019], while the approximation gap to W_2^2 is $O(\varepsilon)$ and vanishes as $\varepsilon \to 0$. We therefore treat Sinkhorn as a calibration proxy rather than part of the law itself.

Corollary 2.6 (EWMA Analogue). *For exponential weights $w_j = \alpha(1-\alpha)^j$, the effective lag is $\tau_\alpha = (1-\alpha)/\alpha$ and the effective sample size scales as $n_{\text{eff}} \asymp 1/\alpha$. Balancing the same statistical and staleness terms therefore yields the same cube-root exponent in the effective memory scale.*

Definition 2.7 (Effective Memory Age). Let an estimator place weights $w_0, \ldots, w_{n-1}$ on the most recent observations, with $\sum_j w_j = 1$. Its *effective memory age* is

$$A(w) \;=\; \sum_{j=0}^{n-1} j\, w_j,$$

and its corresponding freshness proxy is the monotone transform $F(w) = 1/(1 + A(w))$. The role of $F(w)$ is purely geometric: it records how old the retained evidence is before any local drift scale is specified. For a uniform window, $A(w) = (n-1)/2$.

Definition 2.8 (Temporal Coherence). For an estimator with weights w at time t, define its *operational staleness*

$$S_t(w) \;=\; \sum_{j=0}^{n-1} w_j\, W_2(\mathcal{P}^*(S, t-j), \mathcal{P}^*(S, t))\,.$$

Its *temporal coherence* is the inverse alignment score

$$C_t(w) \;=\; \frac{1}{1 + S_t(w)}.$$

If the target path is locally W_2-Lipschitz at rate ζ_t, in the sense that $W_2(\mathcal{P}^*(S, t-j), \mathcal{P}^*(S, t)) \le \zeta_t j$ for the relevant lags, then

$$S_t(w) \;\le\; \zeta_t A(w), \qquad C_t(w) \;\ge\; \frac{1}{1 + \zeta_t A(w)}.$$

In that sense, $F(w)$ is the rate-free freshness proxy, while $C_t(w)$ measures operational alignment with the current distribution.

The quantity $A(w)$ gives a compact description of the temporal geometry of a finite-memory estimator. Short windows preserve temporal coherence at the cost of variance; long windows reduce variance at the cost of staleness. The cube-root horizon marks the point where that exchange stops improving. For a uniform window under approximately constant local drift, the induced staleness scales as $S_t(w) \approx \zeta(n-1)/2$, which is the same age term that appears in theorem 2.1 up to constants.

Proposition 2.9 (Finite-Memory Floor via Gaussian Location Subclass)**.** *Let $\mathcal{P}_{\zeta,m}$ be the class of length-m distribution paths for which the following hold:*

1. *the path is W_2-Lipschitz with rate at most ζ;*
2. *the lower-bound construction is carried out on the Gaussian location subclass $X_t \sim \mathcal{N}(\mu_t, \sigma^2 I_d)$ with $\|\mu_t - \mu_s\|_2 \leq \zeta|t-s|$;*
3. *the estimator at time m only uses the last m samples.*

Measure risk by the endpoint W_2 error, which coincides with Euclidean endpoint error in the Gaussian location subclass. Then, in the critical-window regime, the minimax tracking error over that class cannot beat the same variance-plus-staleness scaling:

$$\inf_{\hat{\mathcal{P}}_m} \sup_{\mathcal{P}^*} \mathbb{E}\, d\big(\hat{\mathcal{P}}_m, \mathcal{P}^*\big) \;\gtrsim\; \max\big\{C_K\, m^{-1/2},\; c\,\sigma^{2/3}\zeta^{1/3}\big\}\,.$$

Consequently, the cube-root horizon is not merely the optimum of one estimator; it is the structural floor of finite memory on this drift class.

Proof. Write $R_m^* := \inf_{\delta_m} R_m(\delta_m)$. We exhibit two independent obstructions. The first is the ordinary finite-sample estimation floor, obtained by restricting to the constant-path subclass $\mu_s \equiv \theta$, which is contained in the admissible drift class. For the two hypotheses $\theta_\pm = \pm\sigma/(2\sqrt{m})$, the induced product-Gaussian laws have Kullback–Leibler divergence

$$\mathrm{KL}(P_+, P_-) = \frac{1}{2\sigma^2}\sum_{j=1}^{m}(\theta_+ - \theta_-)^2 = \frac{1}{2\sigma^2} m \left(\frac{\sigma}{\sqrt{m}}\right)^2 = \frac{1}{2}.$$

For any estimator δ_m, the plug-in test $\psi = \mathbf{1}\{\delta_m \geq 0\}$ must err whenever the estimate falls on the wrong side of the origin; on that event the endpoint error is at least $\sigma/(2\sqrt{m})$. By Le Cam's two-point lemma and Pinsker's inequality, the average testing error is at least

$$\frac{1 - \mathrm{TV}(P_+, P_-)}{2} \geq \frac{1 - \sqrt{\mathrm{KL}(P_+, P_-)/2}}{2} \geq \frac{1}{4}.$$

Hence

$$R_m^* \;\geq\; \frac{\sigma}{8\sqrt{m}},$$

so one valid explicit choice is $c_0 = 1/8$.

The second obstruction is local drift. Fix any $1 \leq h \leq m$ and define

$$\beta_h := \min\Big\{\zeta,\; \frac{\sigma}{4h^{3/2}}\Big\}, \qquad \mu_{t-j}^{\pm,h} := \pm\,\beta_h\,(h-j)_+, \qquad 0 \leq j \leq m-1,$$

where $(u)_+ := \max\{u, 0\}$. Each path belongs to the admissible class because its slope magnitude is at most $\beta_h \le \zeta$. The endpoint gap is

$$|\mu_t^{+,h} - \mu_t^{-,h}| = 2\beta_h h.$$

The corresponding Gaussian product laws satisfy

$$\mathrm{KL}(P_{+,h}, P_{-,h}) = \frac{1}{2\sigma^2} \sum_{j=0}^{m-1} \big(2\beta_h (h-j)_+\big)^2 \le \frac{2\beta_h^2}{\sigma^2} \sum_{r=1}^{h} r^2 \le \frac{2\beta_h^2}{\sigma^2} h^3 \le \frac{1}{8}.$$

Applying the same reduction from estimation to testing, now with endpoint error threshold $\beta_h h$, and using Le Cam together with Pinsker gives average testing error at least

$$\frac{1 - \mathrm{TV}(P_{+,h}, P_{-,h})}{2} \ge \frac{1 - \sqrt{\mathrm{KL}(P_{+,h}, P_{-,h})/2}}{2} \ge \frac{3}{8}.$$

Therefore

$$R_m^* \;\ge\; \frac{3}{8}\,\beta_h h \;\ge\; \frac{3}{32}\,\min\{\zeta h,\ \sigma h^{-1/2}\}$$

which is again a fully explicit numerical lower bound. Since h was arbitrary,

$$R_m^* \;\ge\; \frac{3}{32} \sup_{1 \le h \le m} \min\{\zeta h,\ \sigma h^{-1/2}\}.$$

Taking the maximum of the constant-path and ramp bounds proves the displayed lower bound. For the cube-root regime, assume $1 \le h_\star := (\sigma/\zeta)^{2/3} \le m$. Choose an integer $\bar{h} \in [h_\star/2, h_\star]$. Then

$$\zeta \bar{h} \ge \frac{1}{2}\,\zeta h_\star = \frac{1}{2}\,\sigma^{2/3}\zeta^{1/3},$$

and

$$\sigma \bar{h}^{-1/2} \ge \sigma h_\star^{-1/2} = \sigma^{2/3}\zeta^{1/3}.$$

Therefore

$$\min\{\sigma \bar{h}^{-1/2},\ \zeta \bar{h}\} \ge \frac{1}{2}\,\sigma^{2/3}\zeta^{1/3},$$

and the ramp bound gives

$$R_m^* \ge \frac{3}{64}\,\sigma^{2/3}\zeta^{1/3}.$$

So one admissible explicit constant in the critical-window regime is $c = 3/64$.

Remark 2.10 (Scope of the lower bound). The lower bound is parametric only in the testing construction: it uses a Gaussian location subclass whose mean path is itself W_2-Lipschitz, and the distance in the lower bound is the endpoint W_2 error (Euclidean on that subclass). That is enough to make the result minimax for any larger class containing that subclass, including the ambient W_2-Lipschitz path class. So the proposition is not claiming that every path is Gaussian; it is claiming that the worst case already appears inside the Gaussian mean-shift subproblem. Equivalently, since the Gaussian location family is a subfamily of the ambient W_2-Lipschitz class at the same drift rate ζ, any lower bound proven on that subclass is automatically a lower bound for the larger class.

Remark 2.11 (Measurement layer). Debiased Sinkhorn divergence serves as a practical measurement layer when the underlying geometry must be estimated from samples. That computational choice supports the tracking law, but the law itself is about the horizon of useful memory, not about any particular OT solver.

3 Empirical Validation

The experiments test a single claim: under continuous drift, memory must be regulated by temporal validity rather than by statistical evidence of abrupt change alone. Empirically, the paper separates three layers that are often collapsed: horizon geometry, horizon tracking, and predictive payoff. The Gaussian sweep and lag–variance frontier validate geometry. The continuous-drift, temporal-validity, and alternating-timescales benchmarks probe tracking. The horizon cost curve assesses predictive payoff. A compact piecewise benchmark is retained only as a phase-wise calibration check.

In the experiments, UMR is a backend-agnostic horizon cap driven by a smoothed local drift proxy. It is applied to fixed-window, EWMA, and multi-horizon backends, and it is also wrapped around ADWIN as one detector-side instance of the same temporal-validity rule.

3.1 Law Validation

We begin with the Gaussian sweep used to validate the finite-memory trade-off. The setup produces the characteristic U-curve: for small windows, error is dominated by variance; for large windows, error is dominated by staleness. The empirical minimum remains finite and moves in the expected direction as drift strengthens, and EWMA behaves as the natural finite-memory analogue rather than as a competing theory.

To make the comparison statistically explicit, we also ran a paired bootstrap over the 20 synthetic seeds (50,000 resamples). Table 1 reports the tail-MAE mean and 95% interval for each method, together with the paired difference relative to the UMR-capped detector instance. The horizon cap remains better than ADWIN, EWMA, and the long fixed window on this benchmark, while the short fixed window remains the strongest baseline.

Table 1: Seed-wise bootstrap summary for the main synthetic benchmark. The first column block reports tail-MAE means with 95% bootstrap intervals across the 20 seeds; the second block reports paired differences versus ADWIN+ UMR.

Method	Tail MAE [95% CI]	Δ vs capped ADWIN [95% CI]
Fixed-100	0.0849 [0.0815, 0.0884]	−0.0130 [−0.0144, −0.0115]
Fixed-500	0.2388 [0.2274, 0.2501]	+0.1409 [+0.1319, +0.1504]
EWMA	0.1292 [0.1263, 0.1319]	+0.0313 [+0.0278, +0.0350]
ADWIN	0.1757 [0.1662, 0.1853]	+0.0779 [+0.0704, +0.0857]
ADWIN + UMR	0.0978 [0.0942, 0.1016]	–

3.2 Lag–Variance Frontier

The U-curve becomes more informative when viewed as a frontier in horizon space. The next figure plots tail error against effective memory horizon for a fixed-window sweep and overlays the operating points of the main methods. The curve is asymmetric: on the left, extra memory still buys visible noise reduction; on the right, extra memory mainly adds age. The knee is where that trade stops paying off.

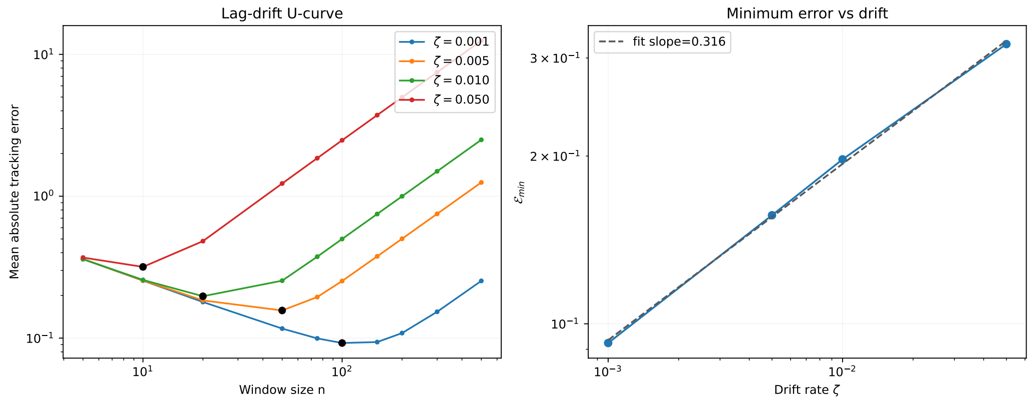

Figure 1: Finite-memory law validation on the Gaussian sweep. Left: tracking error as a function of window size, showing the predicted U-shape for multiple drift rates. Right: the minimum achievable error decreases as drift weakens, consistent with the variance–staleness trade-off. The figure validates the existence of a finite useful-memory optimum, which underlies the rest of the paper.

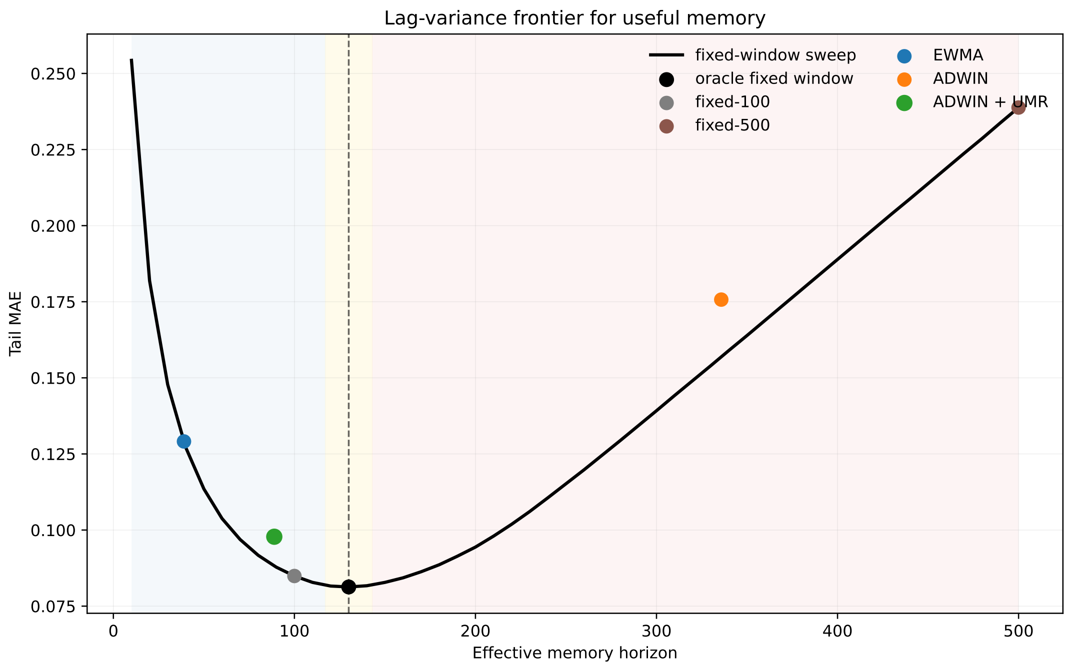

Figure 2: Lag–variance frontier for useful memory. The fixed-window sweep traces the empirical frontier: short horizons are noisy, long horizons are stale, and the knee marks the useful-memory optimum. The frontier is asymmetric because the gain from more memory saturates while the cost of age keeps accumulating. ADWIN sits far to the stale side, whereas ADWIN+ UMR lies close to the frontier knee.

3.3 Over-Memory Under Continuous Drift

The next benchmark focuses on the operational failure mode. We compare five methods on a smooth drifting stream: a short fixed window, a long fixed window, EWMA, ADWIN, and an ADWIN wrapper equipped with the UMR horizon cap. The question is when memory becomes stale before changepoint evidence accumulates.

Table 2: Continuous-drift benchmark under a smooth drift rate. The key gap is between statistical evidence and temporal validity: ADWIN keeps a much wider window than the UMR-capped horizon scale, while the long fixed window collapses under lag.

Method	Tail MAE	Mean horizon	Drift events	Cap events
Fixed-100	0.0849	100.0	–	–
Fixed-500	0.2388	500.0	–	–
EWMA	0.1291	39.0	–	–
ADWIN	0.1757	335.8	3.9	0.0
ADWIN + UMR	0.0978	88.9	0.0	1374.7

Figure 3: Continuous-drift benchmark. Fixed-500 is the stale-memory baseline; ADWIN retains far more memory than the oracle-scale horizon; ADWIN+UMR pulls the effective horizon toward the useful-memory scale and exposes a 'cap-only' regime, where memory becomes operationally stale before statistical changepoint evidence appears.

The key phenomenon is the separation between cap activation and the Hoeffding test. The cap can activate while the test remains silent, which makes memory validity distinct from the evidence threshold for abrupt change.

3.4 Temporal Validity Gap

The gap becomes more explicit when the drift rate varies. The next figure plots the first UMR cap activation for the detector wrapper and the first changepoint decision of ADWIN as the drift rate changes. In the smooth-drift regime, the cap activates much earlier than the detector; as the drift becomes faster, the gap shrinks but does not disappear.

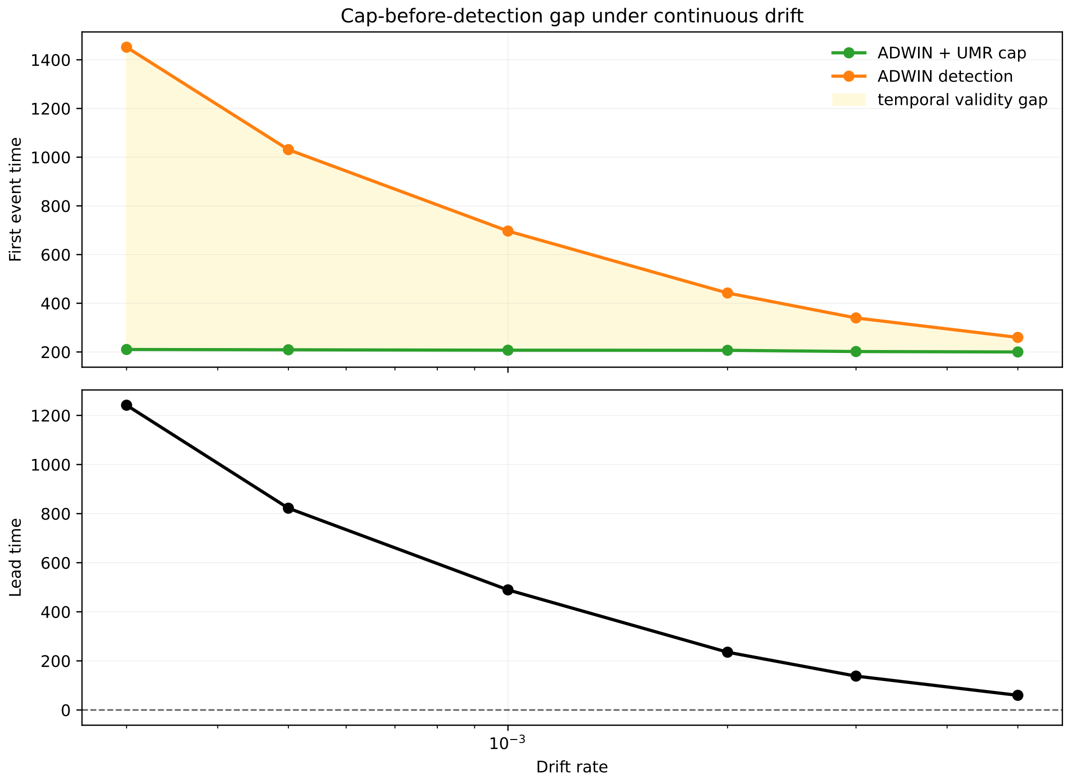

Figure 4: Temporal validity gap across drift rates. The top panel compares the first ADWIN+ UMR cap activation with the first ADWIN detection. The bottom panel shows the resulting lead time. Under slow drift, memory becomes operationally stale long before changepoint evidence is available; under faster drift, the gap narrows but remains positive.

3.5 Fixed-Horizon Instability and Recovery Asymmetry

The next benchmark makes fixed-horizon instability visible. The stream alternates between slow drift, fast bursts, recovery plateaus, and micro-bursts with irregular durations, so the local oracle horizon need not be globally stable. In that setting, a single fixed window can look reasonable on average while remaining locally misaligned for long stretches. This benchmark separates three questions that are often conflated: whether the oracle horizon is time-varying, whether an online regulator can track that changing scale, and how far each method remains from the local oracle horizon over time.

This benchmark makes two things visible. First, horizon validity can vary even when aggregate prediction error stays competitive. Second, the adjustment is asym-

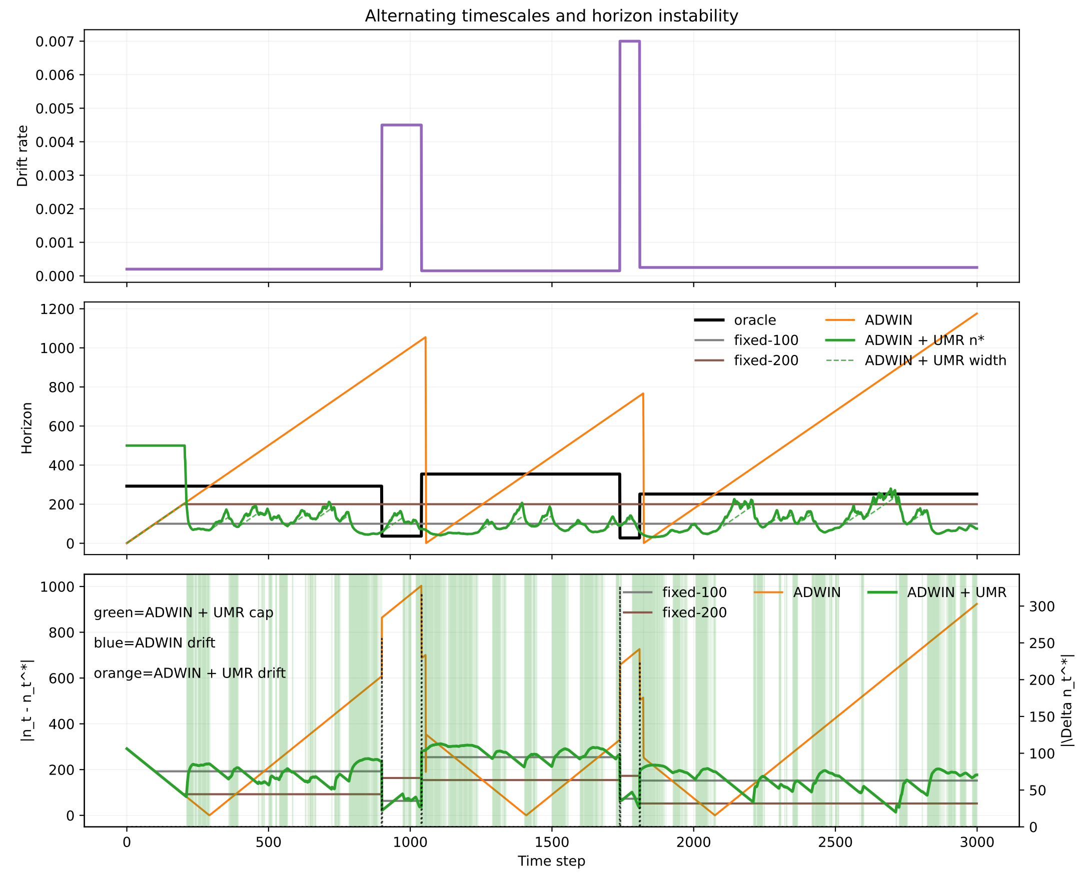

Figure 5: Alternating-timescales instability. The top panel shows the drift schedule, the middle panel compares the local oracle horizon with fixed windows and ADWIN+ UMR, and the bottom panel shows horizon regret over time. The figure exposes local horizon instability under regime changes.

metric: contraction of the useful horizon is easier to trigger than re-expansion after a low-drift stretch. The pattern is operational evidence of path dependence in temporal memory adaptation. Proposition 3.1 formalizes the same asymmetry in the idealized EMA-to-horizon map.

This asymmetry reflects the temporal update dynamics of EMA-driven regulators: contraction is easier to trigger than re-expansion after a low-drift stretch.

Proposition 3.1 (Asymmetric Horizon Recovery)**.** *Let the drift estimator satisfy the EMA recursion*

$$\hat{\zeta}_t \;=\; \alpha\,\Delta_t + q\,\hat{\zeta}_{t-1}, \qquad q := 1-\alpha \in (0,1),$$

with $\Delta_t = \delta_{\text{slow}}$ *in the slow regime and* $\Delta_t = M$ *in the burst regime. Assume the horizon map is unclipped on* $[\delta_{\text{slow}}, M]$ *so that*

$$n^*(\delta) \;=\; \left(\frac{C_K}{\delta}\right)^{2/3}$$

throughout the transition. Write $n_b := n^*(M)$ *and* $n_s := n^*(\delta_{\text{slow}})$*. For any* $0 < \varepsilon < n_s - n_b$*, define the inverse thresholds*

$$\delta_b(\varepsilon) := C_K\,(n_b+\varepsilon)^{-3/2}, \qquad \delta_s(\varepsilon) := C_K\,(n_s-\varepsilon)^{-3/2}.$$

Then the exact recovery time from the slow regime to the burst regime is

$$T_{\text{contract}}(\varepsilon) \;=\; \left\lceil \frac{\log\!\big((M-\delta_{\text{slow}})/(M-\delta_b(\varepsilon))\big)}{\log(1/q)} \right\rceil,$$

while, after a burst of duration T_{burst} steps, the exact recovery time back to the slow regime is

$$T_{\text{expand}}(\varepsilon) \;=\; \left\lceil \frac{\log\!\big((\hat{\zeta}_{\text{end}}-\delta_{\text{slow}})/(\delta_s(\varepsilon)-\delta_{\text{slow}})\big)}{\log(1/q)} \right\rceil,$$

where

$$\hat{\zeta}_{\text{end}} \;=\; M-(M-\delta_{\text{slow}})q^{T_{\text{burst}}}$$

is the burst-end drift estimate. If

$$T_{\text{burst}} \;\geq\; T_0(\varepsilon) \;:=\; \left\lceil \frac{\log(1-\rho_\varepsilon)}{\log q} \right\rceil, \qquad \rho_\varepsilon := \frac{\delta_s(\varepsilon)-\delta_{\text{slow}}}{M-\delta_b(\varepsilon)},$$

then $T_{\text{expand}}(\varepsilon) > T_{\text{contract}}(\varepsilon)$. Moreover, for small ε,

$$M-\delta_b(\varepsilon) = \tfrac{3}{2}C_K n_b^{-5/2}\,\varepsilon + O(\varepsilon^2), \qquad \delta_s(\varepsilon)-\delta_{\text{slow}} = \tfrac{3}{2}C_K n_s^{-5/2}\,\varepsilon + O(\varepsilon^2),$$

so the slow-side threshold distance is smaller by a factor $\big(\delta_{\text{slow}}/M\big)^{5/3}(1+o(1))$.

Proof. The EMA recursion has the closed form

$$\hat{\zeta}_{t_0+k} = \Delta^* + \big(\hat{\zeta}_{t_0}-\Delta^*\big)q^k$$

whenever the regime is constant with fixed point Δ^*. Hence after the transition from δ_{slow} to M,

$$\hat{\zeta}_{t_0+k} = M-(M-\delta_{\text{slow}})q^k,$$

and after the return to the slow regime,

$$\hat{\zeta}_{t_0+T_{\text{burst}}+k} = \delta_{\text{slow}} + (\hat{\zeta}_{\text{end}}-\delta_{\text{slow}})q^k.$$

Because $n^*(\delta)$ is monotone decreasing on the unclipped interval and its inverse is $g(n) = C_K n^{-3/2}$, the condition $|n^*(\hat{\zeta}_{t_0+k})-n_b| \leq \varepsilon$ is equivalent to $\hat{\zeta}_{t_0+k} \geq \delta_b(\varepsilon)$, which gives the displayed formula for $T_{\text{contract}}(\varepsilon)$. Likewise, $|n^*(\hat{\zeta}_{t_0+T_{\text{burst}}+k})-n_s| \leq \varepsilon$ is equivalent to $\hat{\zeta}_{t_0+T_{\text{burst}}+k} \leq \delta_s(\varepsilon)$, which yields $T_{\text{expand}}(\varepsilon)$.

The inequality $T_{\text{expand}}(\varepsilon) > T_{\text{contract}}(\varepsilon)$ holds whenever the burst lasts long enough that

$$1-q^{T_{\text{burst}}} \geq \rho_\varepsilon,$$

since this makes the numerator of T_{expand} larger than that of T_{contract} by the exact threshold relation above. The small-ε expansions follow by Taylor expansion of $n \mapsto C_K n^{-3/2}$ around n_b and n_s. □

Remark 3.2 (Effect of clipping). If clipping at $n_{\min}$ is active near the burst side, then contraction can only be faster. The observed asymmetry is therefore conservative: clipping weakens the theoretical barrier in the burst regime, but it cannot eliminate the longer slow- side recovery. In streams with gradual bursts, the observed ratio can exceed the step-burst threshold because $\hat{\zeta}_{\text{end}}$ does not reach M.

Table 3: Mean horizon regret over the first 50 steps after each regime change. The transition-window ablation shows that contraction and re-expansion are not symmetrical. Lower is better.

Method	Contraction	Expansion
fixed-100	63.3	254.2
fixed-200	163.3	154.2
ADWIN	888.8	435.8
ADWIN + UMR	58.4	307.7

3.6 Drift-EMA Sensitivity

The recovery asymmetry persists when the drift estimator itself is varied. The next sweep changes the EMA coefficient α in the drift estimator while keeping the alternating-timescales benchmark fixed. Larger α makes the drift estimate more reactive, reducing contraction regret while expansion regret stays persistently larger; the expansion-to-contraction ratio remains above one throughout the sweep and is higher at the larger α values. Appendix D then shows that this sensitivity can be reduced by an observable prefix-validation study for (d, α), which shifts the operating point of the regulator and helps interpret the sensitivity of the drift proxy. To keep the closed-form claims auditable, the repository also includes a short numerical check for Proposition 2.8, Proposition 3.1, and Theorem 2.1 against the simulation code used in the paper. The same appendix quantifies the local calibration cost of misestimating C_K: near the optimum, the excess tracking error is quadratic in relative calibration error, so moderate bias is tolerable while large bias moves the operating horizon materially.

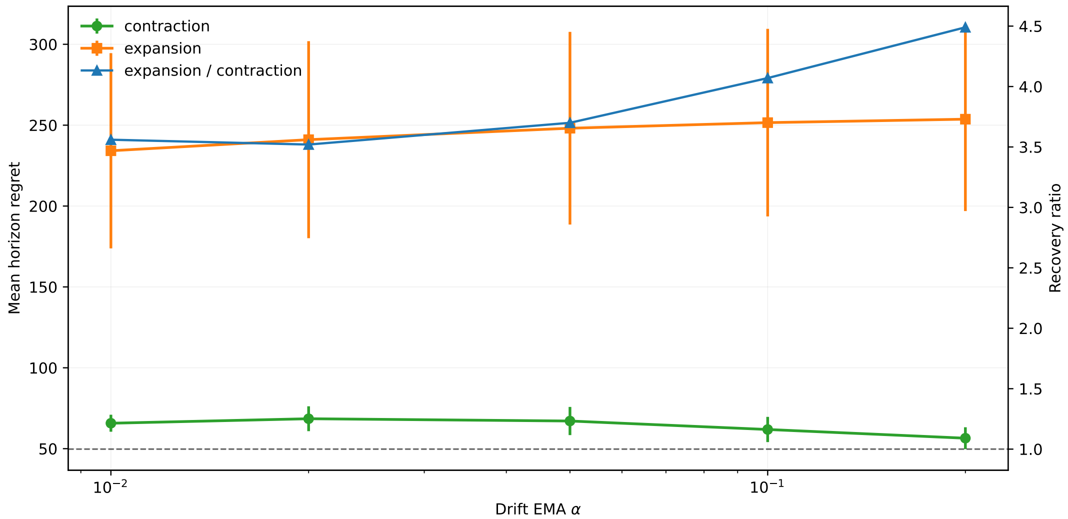

Figure 6: EMA sensitivity of horizon recovery on alternating timescales. The drift-EMA coefficient changes how quickly the regulator reacts to bursts, but the asymmetry remains: expansion regret stays above contraction regret across the sweep, and the ratio is larger at higher α in this benchmark.

3.7 Horizon Cost Curve

The next question is when horizon misalignment becomes operationally expensive. For each backend, we compare its instantaneous horizon n_t with the local cube-root target n_t^* and plot the excess error $E_t - E_t^\star$ against both the absolute gap $|n_t - n_t^*|$ and the relative gap $|\log(n_t/n_t^*)|$. Here $E_t^\star$ is the best error achieved at time t by the fixed-window oracle grid, and the curve is binned over all seeds and time points.

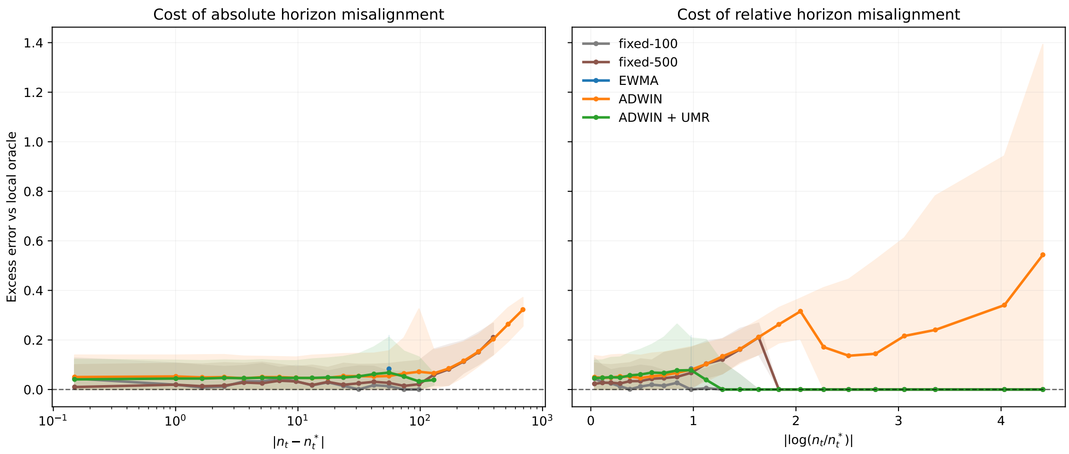

Figure 7: Excess error as a function of horizon misalignment. The relation is backend dependent rather than universal: fixed-100 stays inside a low-cost band, whereas fixed-500 and ADWIN pay substantially more once the gap widens. ADWIN+ UMR sits between these extremes. Some curves terminate early because the corresponding method has no observations beyond that gap range; the visible endpoints are data-limited, not clipped.

The main lesson is that the paper exposes a measurable tolerance band around the useful-memory scale, together with backend-specific degradation once that band is exceeded.

3.8 Phase-Wise Recovery Check

The piecewise-drift benchmark changes the local drift rate in three phases, and the adaptive horizon is compared with the offline oracle fixed window for each phase. The calibration is held fixed across phases; only the stream changes. The detailed calibration check is reported in appendix B. In the main text, the point is simply that one calibration remains in the right operational range when the drift regime changes discretely.

3.9 Representative Real-Stream Case Study

The synthetic experiments isolate the geometry of horizon control and carry the main claim of the paper. The public-stream evidence is therefore kept light: ELEC2 remains the representative case study, while a short Bikes summary is provided in appendix E. The main point is simply that coarse horizon adaptation still matters on public data, but the payoff remains backend dependent.

3.10 Controlled Downstream Check

The downstream classification benchmark is reported in Appendix F. Its role is purely mechanistic: it checks whether the cap-only regime carries a measurable predictive cost when the backend is a windowed classifier rather than a drift detector. The effect is directionally consistent with the main claim but modest in magnitude, so it is a supporting check rather than a dominant result.

3.11 What the Figures Show

Taken together, the figures support three claims. First, the cube-root law is a real geometric law for useful memory rather than a fitting artifact. Second, horizon tracking is distinct from changepoint detection: memory can become stale before statistical evidence is strong enough to declare a change. Third, horizon control has operational value, but its predictive payoff depends on the backend and on the drift regime. The ELEC2 case study reinforces the same temporal-validity gap on public data, but the effect is modest relative to the main synthetic result.

4 Related Work

Tracking under nonstationarity. Dynamic regret, online least-squares, and adaptive filtering ask how much past evidence should remain relevant once the target is moving [Besbes et al., 2015, Cheung et al., 2019, Yuan and Lamperski, 2020, Mazzetto and Upfal, 2023]. Here the same question is kept in distributional form, with the memory horizon as the main object of analysis.

Variation budgets and horizon selection. The $\zeta^{1/3}$ exponent also appears in non-stationary stochastic optimization and bandits [Besbes et al., 2015, Cheung et al., 2019, Wang et al., 2024, Yuan and Lamperski, 2020]. Those results concern gradient-based methods on convex losses in Euclidean space. Here, drift is W_2-Lipschitz on measures, the estimator class is causal averaging kernels, and Proposition 2.9 makes the $\zeta^{1/3}$ rate a minimax lower bound for arbitrary window-restricted estimators.

The same cube-root tuning shows up in sliding-window analyses of drifting data [Bifet and Gavaldà, 2007]: the window length is the control variable, but the quantity being protected is temporal coherence rather than regret alone. The same perspective appears here, but with the geometry changed from scalar losses to distributional tracking.

Jiang, Li, and Zhang study online stochastic optimization under a Wasserstein nonstationarity measure and obtain distribution-aware regret guarantees [Jiang et al., 2020]. Keehan, Anderson, and Wiesemann analyze a related drift-aware weighting problem in Wasserstein distributionally robust optimization [Keehan et al., 2025]. The present focus is narrower: when memory itself is finite, how long can retained evidence remain useful before age becomes the dominant source of error?

Transport geometry. Wasserstein distance and Sinkhorn-type estimators are natural because they respect the geometry of the signal space [Genevay et al., 2019, Niles-Weed and Bach, 2019, Goldfeld et al., 2024, Akyildiz et al., 2024]. In our setting, geometry quantifies how much old evidence can be retained before it becomes stale.

Adaptive forgetting and detection. EWMA, forgetting-factor RLS, and Kalman filtering all balance variance reduction against responsiveness. The cube-root law makes temporal validity explicit. The asymmetry result is adjacent to changepoint detection and EMA-based control, but the contribution here is the gap between those questions: once drift evidence is mapped into a horizon rule, re-expansion can be slower than contraction.

This also connects to the concept-drift literature more broadly [Gama et al., 2004, 2014]. In particular, the "Window Dilemma" argument of Hinder et al. [2023] emphasizes that drift detection is ill-posed when the window scale is itself uncertain. The same issue appears here in a different form: temporal validity is not fixed in advance, so the horizon becomes part of the object that must be estimated.

Public-stream baselines. Window-Dilemma-style and MELO-style baselines widen the empirical comparison. The comparison is broader, but the claim stays the same: the cube-root regulator is a worst-case horizon law for the class studied here, not a claim about every possible drift-path class.

Multi-horizon ensembles attack the same practical issue from another angle by maintaining several historical windows and combining their predictions through uncertainty or expert weighting. The public-stream experiments use that family only as a neighboring design point, not as the main object of analysis [Rajput et al., 2025].

Freshness and Age of Information. The Age-of-Information literature asks how old the last received update is and how update policies should balance communication cost against information staleness [Kaul et al., 2012, Yates et al., 2021]. The distributional analogue is the same question under W_2 drift: useful memory has a finite worst-case horizon, and the cube-root law quantifies that worst-case scale for the class studied here.

5 Limitations and Scope

Scope of the law. The cube-root law is exact for the finite-memory averaging class analyzed in this paper, and the lower bound is stated for window-restricted estimators on a W_2-Lipschitz drift class. The lower bound itself is proved through a Gaussian location subclass and endpoint W_2 error, so its structural claim is worst-case and subclass-based rather than a universal theorem over all possible online estimators.

More refined drift classes may admit tighter staleness growth and different horizon exponents. The analysis here isolates the linear-staleness law induced by the W_2-Lipschitz path bound.

Dependence on local drift estimation. The operational regulator depends on a local estimate of drift. The current benchmark suggests that the remaining gap to the oracle comes mostly from drift estimation. At the same time, this is not just a weakness: the paper now shows that a short observable calibration prefix can materially improve the regulator, so drift estimation is part of the method rather than a hidden nuisance parameter. Better local geometry estimators may therefore sharpen horizon recovery without changing the law.

Geometry beyond the current metric. The formulation uses Wasserstein geometry and a debiased Sinkhorn measurement layer when samples must be compared computationally. Other geometries may admit analogous trade-offs, but the constants and the calibration of useful memory may change.

What the experiments show. The current experiments are not a leaderboard over drift detectors. The public stream result is a representative case study. It shows a specific operational distinction: a tracker can still have statistically insufficient changepoint evidence while already being stale in the sense of useful memory.

Scope of the comparison suite. The benchmark suite now includes a Window-Dilemma-style switcher and a MELO-style multi-horizon hedge, but it is still not a full survey of adaptive tracking methods. The analysis remains centered on the worst-case cube-root law and on the temporal-validity interpretation of memory.

The drift proxy itself has a practical resolution limit: very small block sizes can grossly overestimate $\widehat{\zeta}_t$, so the calibration prefix needs blocks large enough to smooth local noise. In the current sweep, $d = 50$ is acceptable, while $d = 100$ is effectively stable.

On the current ELEC2 and Bikes arenas, adding UMR to the multi-horizon hedge changes results only modestly and in a benchmark-dependent way; it acts as a horizon prior, not a universal replacement for uncertainty weighting.

Path-class refinement. The linear staleness term comes from the local W_2-Lipschitz path bound and is therefore a worst-case statement for that class. Characterizing narrower drift classes in which temporal misalignment accumulates more slowly, together with matching lower bounds, remains open.

6 Conclusion

How long can memory remain useful when the target distribution keeps moving? Under the drift class studied here, useful memory has a finite temporal horizon, and in the worst-case linear-staleness regime that horizon obeys a cube-root law. The governing object is a coherence-lag trade-off: preserving temporally coherent evidence requires limiting the lag induced by old memory.

The theoretical contribution is a decomposition of tracking error into a variance term and a staleness term, together with a lower bound showing that the resulting finite-memory floor is structural. The operational contribution is a horizon regulator: UMR turns the worst-case law into a temporal-validity cap, which keeps the effective horizon from drifting far into stale-memory territory.

The strongest empirical lesson is that horizon geometry, horizon tracking, and predictive payoff are distinct objects. Horizon misalignment carries a nonlinear operational cost, but the transfer from gap to error depends on the backend and on the drift regime.

Empirically, the signature regime is ‘cap-only‘: the memory horizon becomes operationally stale before there is enough statistical evidence for a changepoint alarm. That separation between evidence of change and temporal validity of memory is the main conceptual payoff here. The alternating-timescales benchmark adds a secondary lesson: contraction is easier to induce than re-expansion, so EMA-driven horizon regulators have a recoverability gap.

ELEC2 is the representative real-stream case in the main text, with a short Bikes summary in the appendix. Together they show that the temporal-validity gap is not synthetic-only.

Finite-memory systems are best judged by how long their retained evidence remains current enough to matter. The design consequence is simple: regulate useful memory and track the age at which evidence stops being current.

Useful memory under drift has geometry, but its predictive value is policy- dependent.

Code and reproducibility artifacts are available in the project repository [Parreira, 2026].

Acknowledgements

With gratitude to everyone who contributed ideas, feedback, tools, encouragement, and care along the way. This work stands on the generosity of colleagues, collaborators, open-source maintainers, and the broader research communities that made it possible.

A Backend + UMR Protocol

This appendix gives the concrete protocol used whenever a backend is paired with the useful-memory regulator (UMR). In the experiments, UMR is a shared backend-agnostic horizon cap: it estimates a local drift scale, smooths that estimate with an EMA, and converts it into a runtime horizon bound. The code paths are: `drift/umr.py` for the drift proxy and horizon rule, `experiments/core/horizon_baselines.py` for horizon-capped estimators, `experiments/core/detectors.py` for the ADWIN+UMR reset logic, and `experiments/core/umr_arena.py` for the shared arena composition.

A.1 Shared Regulator

At each time step, the regulator maintains a local drift estimate and a target horizon:

$$\Delta_t = \frac{|\bar{x}_t^{(d)} - \bar{x}_{t-d}^{(d)}|}{d}, \qquad \hat{\zeta}_t = \alpha \Delta_t + (1-\alpha)\hat{\zeta}_{t-1},$$

$$n_t^* = \mathrm{clip}\left(\left(\frac{C_K}{\max(\hat{\zeta}_t, 10^{-9})} \right)^{2/3}, n_{\min}, n_{\max} \right).$$

There is no explicit cooldown parameter. The only stabilization is clipping through $n_{\min}$ and $n_{\max}$, plus the EMA smoothing in $\hat{\zeta}_t$.

A.2 Horizon-Capped Backends

For estimators whose natural state is a window size or effective horizon (fixed-window mean, EWMA, Window-Dilemma-style switching, MELO-style expert mixing), the runtime horizon is simply clipped by the regulator:

$$h_t^{\text{eff}} = \min(h_t^{\text{backend}}, n_t^*).$$

The backend then computes its estimate using h_t^{eff} exactly as it would have used h_t^{backend}.

A.3 Detector Wrapper Instance

One detector-side instantiation uses ADWIN on a scalar signal. The wrapper maintains a rolling buffer of the last $n_{\max}$ observations and applies the same cap logic to the detector width:

1. Update the drift proxy and compute n_t^*.
2. Feed the new observation to ADWIN.
3. Read the current ADWIN width w_t.
4. If $w_t > n_t^*$, record a cap event, reset ADWIN, and re-feed the most recent n_t^* buffered observations to the fresh detector.
5. If the detector then reports drift, record a warning.

The cap event is therefore a horizon event, not a detector event. The detector warning is still ADWIN's own decision after the reset path. This wrapper is one instance of the shared UMR rule, not the defining form of the regulator.

A.4 Hyperparameters

Three parameter families are used.

Synthetic core suite.

Real-stream arenas. The ELEC2 and Bikes arenas use the shared arena builder in `experiments/core/umr_arena.py` with:

- `fixed_window` = 100, `fixed_short_window` = 50, `fixed_long_window` = 300;
- `ewma_alpha` = 0.05;
- `block_size` = 24, `ema_alpha` = 0.03;

Table 4: Synthetic benchmark defaults used by `run_cuberoot_adwin.py`.

Scenario	Default hyperparameters
Constant	`drift` = 0.001, `fixed_window` = 100, `fixed_long_window` = 500, `ewma_alpha` = 0.05, `adwin_delta` = 0.002, `Ck` = 1.0, `drift_window` = 100
Piecewise	`piecewise_drifts` = $(0.0005, 0.003, 0.001)$ with the same window, EWMA, ADWIN, and C_K settings as above
Alternating	`piecewise_drifts` = $(0.0002, 0.0045, 0.00015, 0.007, 0.00025)$, `piecewise_lengths` = $(900, 140, 700, 70, 1190)$, `fixed_long_window` = 200, and the same remaining defaults
Common	$n_{\min} = 10$, $n_{\max} = 500$

- `min_window` = 30, `max_window` = 300, `scale` = 1.25;
- `baseline_window` = 100, `prefix_length` = 2000;
- `adwin_delta` = 0.03;
- Page-Hinkley event detector: $\delta = 0.01$, threshold 84.0, $\alpha = 0.9999$;
- Window-Dilemma windows $(50, 100, 300)$ with quantiles $(0.33, 0.67)$;
- MELO-style expert windows $(30, 50, 100, 200, 300)$ with learning rate 6.0 and discount 0.995.

Drift-proxy calibration. When C_K is not fixed externally, the code estimates it from a stationary prefix via `calibrate_umr_constant`, i.e. standard deviation times the square root of prefix length. The drift-proxy appendix additionally reports the prefix-validated (d, α) sensitivity study, but that is a robustness study rather than a replacement for the fixed-regulator core.

A.5 Algorithm

The exact implementation details are in the source files listed at the top of this appendix.

B Phase-Wise Recovery Details

This appendix presents the phase-wise calibration check for the UMR-regulated ADWIN benchmark. The result is that the same calibration stays in the right operational range when the drift regime changes discretely.

C Finite-Sample Geometry Check

This appendix records a numerical sanity check for the finite-sample term in lemma 2.2. We estimate empirical W_2 between two i.i.d. samples from the same distribution using exact assignment in low and moderate dimension, then fit a

Algorithm 1. UMR update rule.

1. **Input:** stream $x_{1:T}$, backend B, prefix length m, block size d, EMA coefficient α, bounds $[n_{\min}, n_{\max}]$, constant C_K.
2. For each $t = 1, \dots, T$:
   (a) Update Δ_t from the last two blocks of length d.
   (b) Update $\hat{\zeta}_t = \alpha \Delta_t + (1-\alpha)\hat{\zeta}_{t-1}$.
   (c) Compute $n_t^* = \mathrm{clip}((C_K/\max(\hat{\zeta}_t, 10^{-9}))^{2/3}, n_{\min}, n_{\max})$.
   (d) If B is horizon-based, set the effective horizon to $\min(h_t^B, n_t^*)$ and evaluate the backend with that horizon.
   (e) If B is a detector wrapper, update the detector; if its internal width exceeds n_t^*, reset it and re-feed the last n_t^* buffered observations; then record any detector warning.
3. **Output:** backend estimate, warnings, n_t^*, drift proxy, and cap events.

Figure 8: UMR update rule.

Table 5: Oracle horizon recovery on piecewise drift. The same calibrated cap tracks the oracle scale across regimes: it is close in the fast-drift phase and remains within the right order of magnitude in the smoother phases.

Drift	Oracle window	Regulated n^*	Oracle MAE	ADWIN + UMR MAE
0.0005	200	191.2	0.0741	0.0874
0.003	50	58.1	0.1258	0.1381
0.001	100	89.7	0.0876	0.1033

log–log slope in n. We also check the Sinkhorn proxy in one dimension at a few fixed values of ε.

Table 6: Empirical scaling check for the finite-sample term. The bounded-support W_2 experiment is close to the expected $n^{-1/2}$ rate in low dimension, and the slope degrades as intrinsic dimension increases. The Sinkhorn proxy is stable for fixed ε, with the prefactor changing as ε varies.

Setting	Estimated slope	Comment
W_2, $d = 1$	-0.48	close to $-1/2$
W_2, $d = 2$	-0.44	slower decay
W_2, $d = 4$	-0.28	visibly slower
W_2, $d = 5$	-0.22	visibly slower
Sinkhorn $\varepsilon = 0.50$	-1.34	proxy constant changes
Sinkhorn $\varepsilon = 0.10$	-1.09	proxy constant changes
Sinkhorn $\varepsilon = 0.05$	-0.88	proxy constant changes

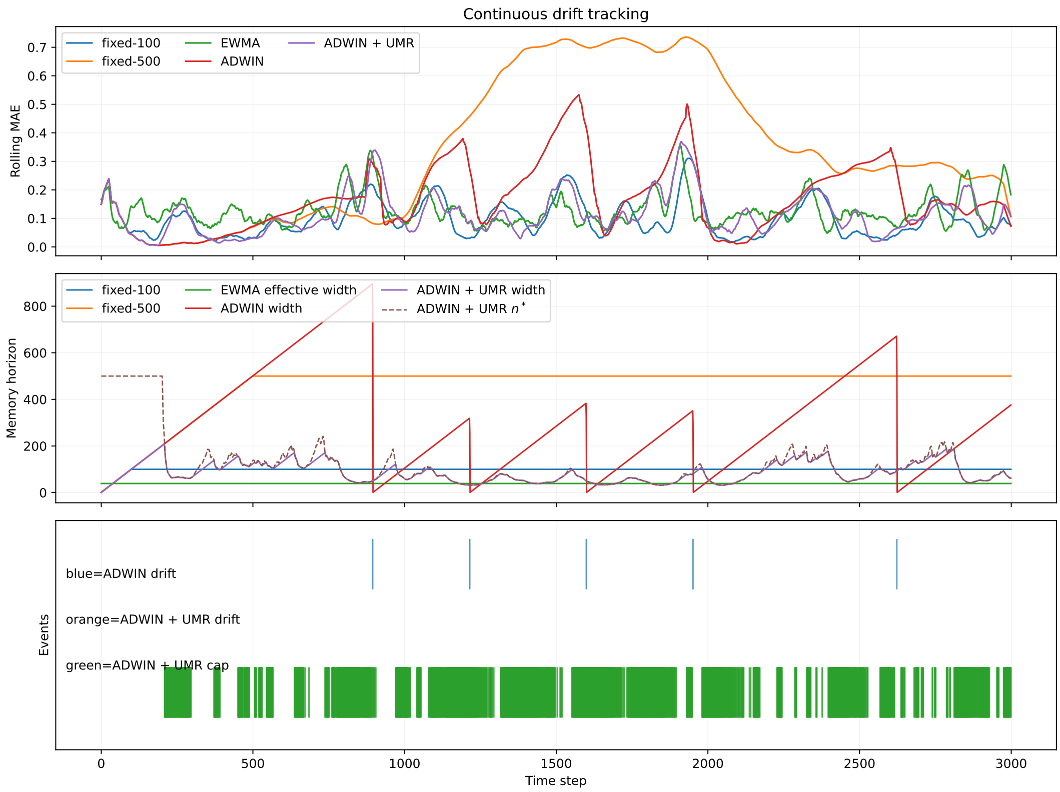

Figure 9: Piecewise-drift recovery. The same regulator calibration follows the phase-dependent oracle window closely enough to preserve the right operational scale in each regime.

The point of this check is not a sharp asymptotic theorem. It is to confirm the modeling choice: the low-dimensional raw-Wasserstein regime is compatible with an $n^{-1/2}$ term, while the Sinkhorn proxy behaves as a fixed-ε measurement layer with a calibration constant that depends on geometry and on regularization.

The Sinkhorn ε ablation shows that fixed-ε bias can substantially inflate the implied horizon constant in high dimension, with the tested $d = 256$ slice ranging from about $1.36\times$ to $4.24\times$ in the linearized horizon factor. The point is not a sharp asymptotic theorem; it is to show that the calibration constant can move materially with regularization even when the exponent structure stays the same.

D Online Drift Estimation and Sensitivity

The dynamic horizon regulator uses a short-window drift proxy built from block differences. For block length d, let $\bar{x}_t^{(d)}$ denote the mean of the most recent block and $\bar{x}_{t-d}^{(d)}$ the mean of the previous block. The instantaneous drift proxy is

$$\Delta_t = \frac{\left|\bar{x}_t^{(d)} - \bar{x}_{t-d}^{(d)}\right|}{d},$$

and the smoothed estimate follows the EMA recursion

$$\widehat{\zeta}_t = \alpha\,\Delta_t + (1-\alpha)\,\widehat{\zeta}_{t-1}.$$

The regulator then sets

$$n_t^* = \operatorname{clip}\left(\left(\frac{C_K}{\max(\widehat{\zeta}_t, 10^{-9})}\right)^{2/3}, n_{\min}, n_{\max}\right).$$

Table 7 reports the sensitivity sweep already used in the paper. Larger α makes the horizon react faster to bursts, which lowers contraction regret but leaves expansion regret larger; the expansion-to- contraction ratio stays above one throughout.

Table 7: Drift-EMA sensitivity on the alternating-timescales benchmark. The ratio remains above one across the sweep, so re-expansion is slower than contraction for every tested α.

α	Contraction	Expansion	Ratio	Tail MAE
0.01	65.73	234.18	3.56	0.0938
0.02	68.51	241.03	3.52	0.0923
0.05	67.10	248.12	3.70	0.0925
0.10	61.86	251.57	4.07	0.0934
0.20	56.48	253.70	4.49	0.0940

The appendix complements the main-text proposition by giving the concrete estimator used in the benchmarks and the empirical sweep used to check that the asymmetry persists as the drift filter is retuned.

D.1 Calibration Cost of C_K Misestimation

Let the regulator use an estimated constant $\widehat{C}_K = rC_K$ while the true tracking law is governed by C_K. Since the selected horizon is $n^*(\widehat{C}_K) = (\widehat{C}_K/\zeta)^{2/3}$, the excess tracking error relative to the oracle choice is

Proposition D.1 (Calibration Cost of C_K Misestimation)**.**

$$\Delta E(r) = \mathcal{E}(n^*(\widehat{C}_K); C_K, \zeta) - \mathcal{E}(n^*(C_K); C_K, \zeta) = C_K^{2/3}\zeta^{1/3}\left(r^{-1/3} + \tfrac{1}{2}r^{2/3} - \tfrac{3}{2}\right).$$

Moreover, for $r = 1 + \delta$ with $|\delta| \ll 1$,

$$\Delta E(r) = \tfrac{1}{9}E_{\min}\,\delta^2 + O(\delta^3),$$

so small calibration errors are locally quadratic and symmetric to second order.

The practical implication is simple: moderate calibration error in C_K is benign to first order, but large systematic bias still moves the operating point materially. The calibration-prefix machinery in Appendix A and the block-size sensitivity table below quantify how this shows up in practice. A short numerical check of the proposition is available in `scripts/calibration_cost.py`.

Table 8: Local calibration-cost sensitivity for the horizon constant. The excess error is normalized by $E_{\min}$, so the table is scale-free. The proposition gives the local quadratic law around $r = 1$; the table shows the exact curve.

$r = \widehat{C}_K/C_K$	Exact $\Delta E/E_{\min}$	Comment
0.90	0.0018	modest underestimation already moves the horizon
0.95	0.0004	local error is very small near the optimum
1.00	0.0000	oracle calibration
1.05	0.0004	local error is very small near the optimum
1.10	0.0015	moderate overestimation remains mild
1.25	0.0085	larger bias becomes visible
1.50	0.0288	large bias moves the operating point

D.2 Calibration Sensitivity Summary

Two short sensitivity checks are enough to capture the practical behavior of the regulator. First, the Sinkhorn regularization level changes the constant but not the exponent: across the tested values, the observed $\zeta^{1/3}$ slope stays at approximately 0.333, while the implied C_K moves from about σ at large ε to roughly 3.3σ at small ε. Second, the block length must not be too small: $d = 10$ produces a very large positive bias in the drift proxy, $d = 50$ is already usable, and $d = 100$ is effectively stable.

Table 9: Practical sensitivity summary for the drift proxy. The epsilon sweep changes the calibration constant C_K but leaves the cube-root exponent intact; the block-size sweep shows that too-small blocks induce severe positive bias in $\widehat{\zeta}_t$.

Knob	Observed effect	Rule of thumb	Comment
ε	slope ≈ 0.333	exponent unchanged	only C_K moves
$d = 10$	+618% bias in $\widehat{\zeta}_t$	too small	unusable
$d = 50$	+7.8% bias in $\widehat{\zeta}_t$	acceptable	practical
$d = 100$	negligible bias	stable	preferred when feasible

D.3 Robustness of the EMA Drift Proxy

Let the block-difference proxy satisfy $\Delta_t = \zeta_t + \xi_t$, where ξ_t is a zero-mean noise term with $\mathrm{Var}(\xi_t) \le \sigma_\Delta^2$ and the EMA recursion is

$$\widehat{\zeta}_t = \alpha \Delta_t + (1-\alpha)\widehat{\zeta}_{t-1}.$$

If $\zeta_t \equiv \zeta$ is constant, then

$$\mathbb{E}[\widehat{\zeta}_t] = \zeta + (1-\alpha)^t(\widehat{\zeta}_0 - \zeta), \qquad \mathrm{Var}(\widehat{\zeta}_t) \le \frac{\alpha}{2-\alpha}\,\sigma_\Delta^2.$$

So smaller α reduces variance but increases lag, while larger α reacts faster but amplifies noise.

Since the regulator uses

$$n_t^* = \left(\frac{C_K}{\widehat{\zeta}_t}\right)^{2/3},$$

the relative horizon error is first-order sensitive to relative drift-estimation error:

$$\frac{\delta n_t^*}{n^*} \approx -\frac{2}{3}\frac{\delta\widehat{\zeta}_t}{\zeta}.$$

Thus persistent overestimation of ζ induces under-memory, and persistent underestimation induces over-memory. If the proxy has multiplicative bias $\mathbb{E}[\Delta_t] = (1+b)\zeta$ with $b > 0$, then asymptotically

$$\frac{\widehat{n}^*}{n^*} = (1+b)^{-2/3},$$

so a 10% overestimate of drift shrinks the target horizon by about 6.5%.

Table 10: EMA robustness check for a constant drift proxy with additive noise. The empirical mean matches the target drift, the empirical variance matches the closed-form $\alpha/(2-\alpha)$ law, and the implied horizon distortion stays close to one when the proxy is unbiased.

α	$\mathbb{E}[\widehat{\zeta}_t]$	$\mathrm{Var}(\widehat{\zeta}_t)$	Theory	$\mathbb{E}[\widehat{n}^*]/n^*$
0.01	1.0000	0.00031	0.00031	1.0002
0.02	0.9999	0.00063	0.00063	1.0004
0.05	1.0000	0.00160	0.00160	1.0009
0.10	1.0000	0.00329	0.00329	1.0018
0.20	1.0002	0.00694	0.00694	1.0038

For persistent overestimation, the same calculation gives systematic under-memory. For example, with a 10% positive bias the asymptotic horizon ratio is $1.1^{-2/3} \approx 0.94$, and with a 25% bias it is about 0.86.

D.4 Prefix-Validated Auto-Calibration

The operational fix is to choose (d, α) on a short calibration prefix by minimizing an *observable* prequential score over a small grid. For each candidate pair (d, α), run the regulated estimator on the prefix, generate the one-step-ahead forecast $\hat{x}_{t|t-1}^{(d,\alpha)}$, and score it by

$$S(d, \alpha) = \frac{1}{T_0 - 1}\sum_{t=1}^{T_0-1}\left(x_{t+1} - \hat{x}_{t|t-1}^{(d,\alpha)}\right)^2.$$

The selected pair minimizes this prequential MSE on the prefix and is then used for the full run. This is implementable online because it only uses observed data and the regulator's own forecasts.

Table 11: Prefix-validated auto-calibration on the synthetic benchmark. The calibration prefix chooses (d, α) by observable one-step-ahead prequential MSE, then the selected parameters are run on the full stream. This is a sensitivity study: the selected pair shifts the operating point, and on this calibration set it improves tail MAE relative to the fixed default.

Scenario	Mean α	Mean d	Tail MAE (default)	Tail MAE (calibrated)
Constant drift	0.19	50.0	0.2087	0.1224
Alternating drift	0.11	62.5	0.1020	0.0890

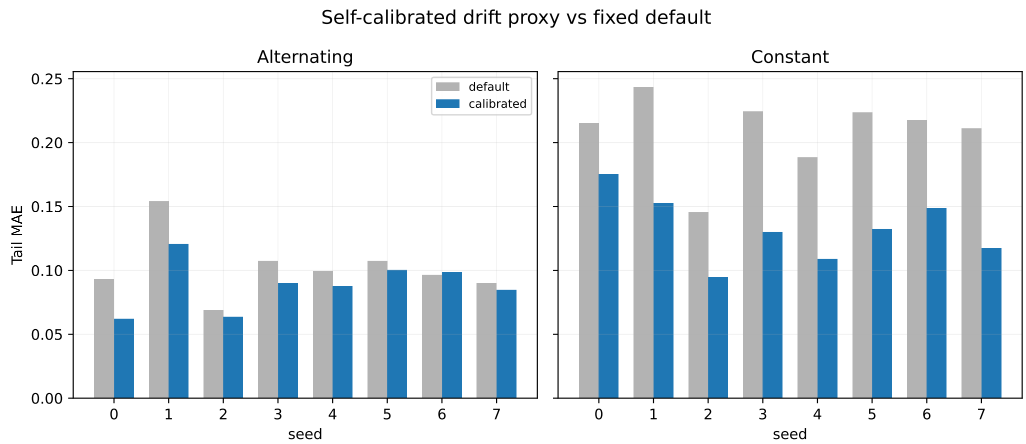

Figure 10: Prefix-validated self-calibration of the drift proxy. The selected (d, α) shifts the operating point of the regulator; on this calibration study it improves tail MAE relative to the fixed default.

The failure mode is also visible: if the proxy is overreactive, the selected window contracts too aggressively and under-memory appears in the tail; if it is too inertial, the horizon lags the drift and over-memory follows. The calibration rule reduces that sensitivity by choosing the grid point that best predicts the next-step observations on the prefix.

The implemented self-calibrating rule is therefore simple: keep a small grid of block scales and EMA coefficients, score each candidate by prequential one-step MSE on a short prefix, then deploy the best candidate for the remainder of the stream. For non-Lipschitz local behavior, the grid should include at least one short and one longer block scale so that bursty changes do not force a single timescale to carry the entire burden. This is an operational refinement, not a replacement for the fixed-regulator results in the main text.

E Additional Public-Stream Summary

ELEC2 is the representative public-stream case in the main text. The only additional summary retained here is the Bikes counterpart, included to show that the same temporal-validity gap is not unique to one stream.

The qualitative picture is the same as in ELEC2: coarse horizon adaptation still matters on public data, but the payoff remains backend-dependent.

Table 12: Bikes public-stream summary. Precision is matched-warning precision against Page-Hinkley events; median lead is the median warning lead in steps.

Method	Precision	Median lead	Leads
EWMA	0.146	132.0	59
EWMA + UMR	0.146	132.0	59
Window Dilemma	0.127	108.5	76
Window Dilemma + UMR	0.128	125.0	76
MELO-style	0.141	147.5	76
MELO-style + UMR	0.129	147.0	71
ADWIN	0.526	92.0	251
ADWIN + UMR	0.520	82.0	141

F Controlled Downstream Check

The controlled synthetic downstream benchmark focuses on the phase-2 cap-only window: UMR has already contracted, while ADWIN remains silent.

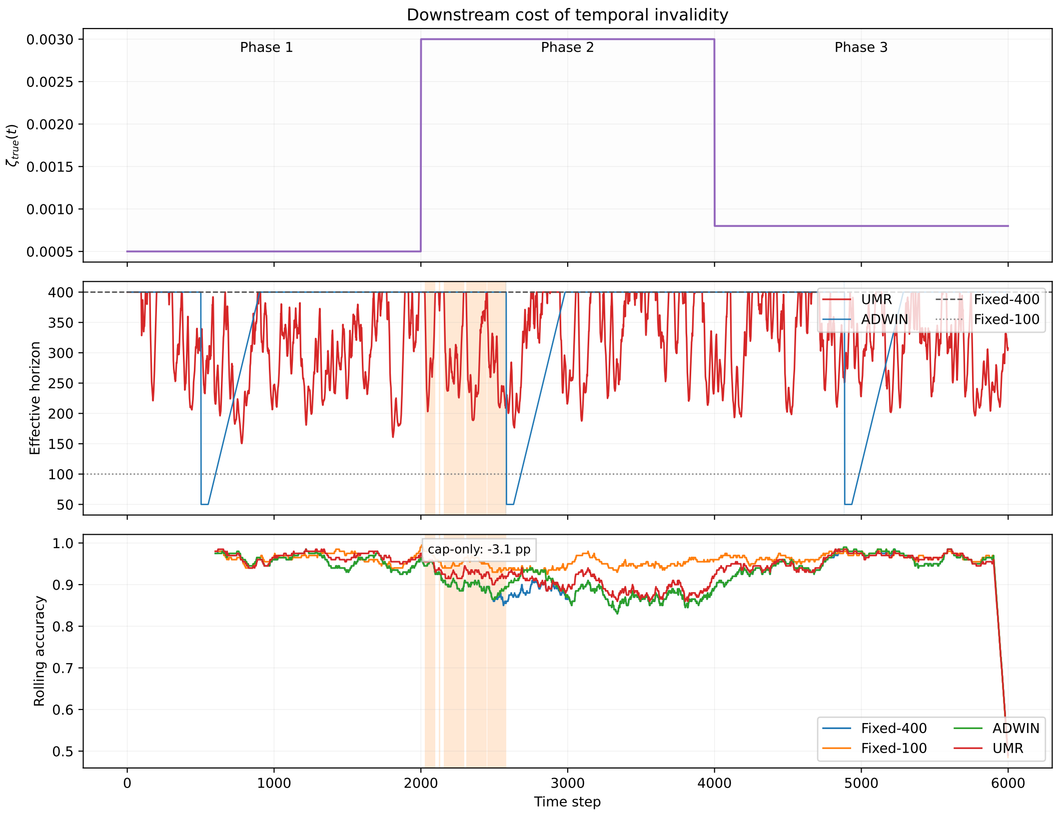

Figure 11: Controlled downstream benchmark. Top: true drift rate. Middle: effective horizons and the cap-only interval. Bottom: rolling prequential accuracy. The representative seed shows a 3.12 point gain for UMR over Fixed-400 within the cap-only window.

Table 13: Controlled downstream benchmark summary. The reported delta is UMR minus Fixed-400 within the cap-only interval, averaged across six seeds. The effect is consistent in direction but modest in magnitude.

Method	Global acc.	Cap-only acc.	Δ cap-only	Mean cap-only width
Fixed-400	0.9304	0.9040	0.00	293.3
Fixed-100	0.9598	0.9576	+5.36	293.3
ADWIN	0.9340	0.9040	0.00	293.3
UMR	0.9415	0.9205	+1.66	293.3

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